

# Week 4 – Quantum Computing

BB84 Protocol, Tensor Product, Multi-Qubit systems

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## Review from Week 3

- Last week, we observed that a single qubit system has the form  $|\psi\rangle = a|\Phi_0\rangle + b|\Phi_1\rangle$ , where  $\{|\Phi_0\rangle, |\Phi_1\rangle\}$  is an orthonormal basis for  $\mathbb{C}^2$ , and where  $a, b \in \mathbb{C}$  with  $|a|^2 + |b|^2 = 1$ .
- We also observed the “Born rule” for measurement, which stated that given  $|\psi\rangle = a|\Phi_0\rangle + b|\Phi_1\rangle$ , the probability of measuring, say  $|\Phi_0\rangle$  is  $|a|^2$ , and if the outcome  $|\Phi_0\rangle$  is observed, then the resulting state after measurement is given by  $\frac{a}{|a|}|\Phi_0\rangle$ .
- The basis  $\{|0\rangle, |1\rangle\}$  is known as the *computational basis*, with  $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ , and  $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ .

Before we move onto studying multi-qubit systems, we will see an application of single qubit systems and measurement in the area of secure information transfer. We first briefly review the classic Vernam cipher (one-time pad) as a motivating step to studying the so-called BB84 (Bennett, Brassard, '84) key exchange protocol.

## BB84 Key Exchange

### Classical Encryption – One-time pad

Suppose Alice wishes to send a message – for purposes of simplicity an  $n$ -bit binary string – to Bob, then the probability that one may “guess” the correct bitstring is  $2^{-n}$ . However, to avoid an eavesdropper Eve from tapping the communication channel and discovering what the message is, Alice may choose to *encrypt* the message before sending it to Bob; Alice and Bob respectively encrypt and decrypt using a secret private key that has already been exchanged between them prior to sending this message. An eavesdropper Eve may still see the encrypted message, but not the actual message. The protocol is illustrated using an example as follows:

1. Alice wishes to send message  $M = 101010111$ , and is using a key  $K = 110110110$  to encrypt it as follows:  $C = M \oplus K = 011100001$ , where  $\oplus$  is the bitwise addition modulo 2 (or the XOR operation). Note that this encryption method requires the key  $K$  to be at least as long as the message.
2. Bob receives  $C$ , and decrypts it using the same key  $M = C \oplus K = 011100001 \oplus 110110110 = 101010111$ .
3. Eve may see  $C$ , but without any knowledge of either  $M$  or  $K$ , their chances of guessing the correct message is still  $2^{-n}$ .

The protocol is thus perfectly secure from an information-theoretic standpoint, under two key assumptions.

- The key is perfectly random.
- The key is used exactly once (hence the name).

Despite these qualities, there are multiple practical limitations to implementing the one-time pad protocol. One of them is secure communication of the private key itself between Alice & Bob. In general, this is an important problem in cryptography, and the BB84 protocol offers an approach using quantum phenomena.

## BB84 – key distribution over a quantum channel

There will be two orthonormal bases that will play a crucial part in this protocol. They are  $Z = \{|0\rangle, |1\rangle\}$ , which is the computational basis, and  $X = \{|+\rangle, |-\rangle\}$ , which is the Hadamard basis.

**Note.** An important thing to note about them is that the two bases are incompatible – a state prepared in one basis, but measured in another, leads to a random measurement outcome.

Let us now examine the basic protocol itself. Alice sends an  $n$ -bit binary string to Bob, hoping that some (sufficiently long) substring can eventually become a private key that the two can use later for secure information transfer, unless they detect intrusion (to be described later). Alice will encode each bit using a randomly chosen basis, using the following encoding scheme.

| Bit \ Basis | $Z$         | $X$         |
|-------------|-------------|-------------|
| 0           | $ 0\rangle$ | $ +\rangle$ |
| 1           | $ 1\rangle$ | $ -\rangle$ |

We first see how Alice and Bob carry out their roles in the key transfer protocol, and later introduce the presence of Eve to show how Alice & Bob may detect Eve’s presence.

1. For each bit, Alice randomly chooses a basis from  $\{X, Z\}$  and sends the corresponding qubit encoding in that basis.
2. For each bit, Bob randomly chooses a basis from  $\{X, Z\}$  and measures the sent qubit in that basis.

After all of the bits have been transferred, Alice and Bob publicly reveal their bases (**not** the bits) for each of the positions. They discard bits in all of the positions where their choice of bases did not match, and retain the others. It is not hard to see that they will end up retaining roughly half of the bits originally sent.

In the positions where their choice of basis did match, they randomly choose  $k$  of the bits to compare publicly. These  $k$  bits will only serve the purpose of detecting the presence of what is called an “intercept-resend”<sup>1</sup> attack carried out by Eve, where she measures Alice’s bit in a randomly chosen basis of her own, then sends her measurement results to Bob. The  $k$  bits used to

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<sup>1</sup>Note that for fundamental quantum mechanical reasons that we will study next week, Eve can not, without avoiding detection, first copy Alice’s qubit, then measure it after bases are revealed.

| Bit # | Alice Bit | Alice Basis | Encoded Qubit | Bob Basis | Bob Bit |
|-------|-----------|-------------|---------------|-----------|---------|
| 1     | 0         | $Z$         | $ 0\rangle$   | $Z$       | 0       |
| 2     | 1         | $X$         | $ -\rangle$   | $Z$       | ?       |
| 3     | 1         | $Z$         | $ 1\rangle$   | $Z$       | 1       |
| 4     | 0         | $X$         | $ +\rangle$   | $X$       | 0       |
| 5     | 1         | $Z$         | $ 1\rangle$   | $X$       | ?       |
| 6     | 0         | $Z$         | $ 0\rangle$   | $Z$       | 0       |
| 7     | 1         | $X$         | $ -\rangle$   | $X$       | 1       |
| 8     | 0         | $X$         | $ +\rangle$   | $Z$       | ?       |
| 9     | 1         | $Z$         | $ 1\rangle$   | $Z$       | 1       |
| 10    | 0         | $X$         | $ +\rangle$   | $X$       | 0       |

Table 1: Alice encodes classical bits into qubit states chosen from two incompatible bases ( $Z$  and  $X$ ). Bob measures each qubit in a randomly chosen basis; outcomes are definite when bases match and indeterminate otherwise.

| Bit # | Sifted Bit | Alice/Bob Basis | Eve Basis | Eve $\rightarrow$ Bob |
|-------|------------|-----------------|-----------|-----------------------|
| 1     | 0          | $Z$             | $Z$       | 0                     |
| 3     | 1          | $Z$             | $X$       | 0/1                   |
| 4     | 0          | $X$             | $X$       | 0                     |
| 6     | 0          | $Z$             | $X$       | 0/1                   |
| 7     | 1          | $X$             | $Z$       | 0/1                   |
| 9     | 1          | $Z$             | $Z$       | 1                     |

Table 2: When Eve measures in the same basis as Alice and Bob, she forwards the correct bit. When she measures in the wrong basis, her outcome is random, and Bob receives either bit with equal probability.

detect this attack will, naturally, not be a part of the final key. The following exercise should help you figure out how Eve's presence may be detected.

**Exercise 1.** *For each of the following questions, assume that we are working only with the qubits in the sifted key positions, i.e. positions where Alice and Bob choose the same basis.*

- For a given key position, what is the probability that Eve chose the same basis as Alice and Bob?*
- What is the probability that Eve measured in the wrong basis? Conditioned on this event, what is the probability that Bob's measurement disagrees with Alice's bit.*
- Compute the probability that a fixed sifted position produces a detection event, i.e. the bit Alice prepared is different from the bit Bob measures.*
- Suppose Alice and Bob publicly compare  $k$  randomly chosen same-basis rounds (a test subset of the sifted key). What is the probability that Eve goes undetected in all  $k$  tests?*

(e) How large must  $k$  be to make the probability of Eve getting detected larger than 99%? <sup>2</sup>

**Conceptual Summary.** BB84 achieves secure key distribution using single-qubit states and quantum measurement. Alice encodes each bit using a qubit prepared in one of two incompatible bases, chosen uniformly at random, and Bob measures each qubit in one of two bases (chosen independently, and uniformly at random). After publicly comparing bases, they keep bits in only the same-basis rounds to form a sifted key. Any eavesdropper must correctly guess the measurement basis, and incorrect guesses disturb the qubits, producing detectable errors in the sifted key; testing a random subset therefore reveals eavesdropping with high probability.

## Multi-qubit systems

Having seen single qubit systems, we now initiate a study of multi-qubit systems, which leads us to the all-important definition of the tensor product. We begin by stating the following fourth postulate.

**Definition 1** (Postulate 4, Nielsen–Chuang). *The state space of a composite physical system is the **tensor product** of the state spaces of the component physical systems. For example, for systems  $|\psi_1\rangle, \dots, |\psi_n\rangle$ , the joint state of the total system is given by  $|\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_n\rangle$ .*

An operational definition of the tensor product for vectors and matrices is provided below, with one example for each.

**Definition 2** (Tensor product of two vectors). *Let*

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad |\phi\rangle = \begin{pmatrix} \gamma \\ \delta \end{pmatrix}.$$

*The tensor product is defined as*

$$|\psi\rangle \otimes |\phi\rangle = \begin{pmatrix} \alpha|\phi\rangle \\ \beta|\phi\rangle \end{pmatrix}.$$

*Writing this out explicitly,*

$$|\psi\rangle \otimes |\phi\rangle = \begin{pmatrix} \alpha \begin{pmatrix} \gamma \\ \delta \end{pmatrix} \\ \beta \begin{pmatrix} \gamma \\ \delta \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix}.$$

**Example 1.**

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \otimes \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \begin{pmatrix} 3 \\ 4 \end{pmatrix} \\ 2 \begin{pmatrix} 3 \\ 4 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 6 \\ 8 \end{pmatrix}.$$

**Definition 3.** *Let*

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}.$$

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<sup>2</sup>To detect an “intercept-resend” attack, Alice begins with  $\approx 2(n+k)$  bits, where  $n$  is the target key length.

The tensor product is defined blockwise as

$$A \otimes B = \begin{pmatrix} aB & bB \\ cB & dB \end{pmatrix}.$$

Writing this explicitly,

$$A \otimes B = \begin{pmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{pmatrix}.$$

**Example 2.** Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

$$A \otimes B = \begin{pmatrix} 1 \cdot B & 2 \cdot B \\ 3 \cdot B & 4 \cdot B \end{pmatrix}.$$

$$A \otimes B = \begin{pmatrix} 1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ 3 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 4 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{pmatrix}.$$

$$A \otimes B = \begin{pmatrix} 0 & 1 & 0 & 2 \\ 1 & 0 & 2 & 0 \\ 0 & 3 & 0 & 4 \\ 3 & 0 & 4 & 0 \end{pmatrix}.$$

**Remark 1.** The constructions above give an *operational* definition of the tensor product using  $2 \times 1$  vectors and  $2 \times 2$  matrices for simplicity; the same block-scaling rule applies to higher-dimensional vectors and matrices, with dimensions multiplying accordingly.

Before we study some of the important properties of the tensor product, note that we use the notations  $|\psi_1\rangle \otimes |\psi_2\rangle = |\psi_1\rangle|\psi_2\rangle = |\psi_1\psi_2\rangle$  interchangeably, depending on convenience.

**Definition 4** (Properties of Tensor Product). *The following is not an exhaustive list, but contains most of the properties most commonly encountered in our study of quantum circuits and algorithms.*

1. **Associativity.**  $(A \otimes B) \otimes C = A \otimes (B \otimes C)$
2. **Non-Commutativity.**  $A \otimes B \neq B \otimes A$ .
3. **Scalar Multiplication.**  $a|\psi_1\rangle \otimes |\psi_2\rangle = |\psi_1\rangle \otimes a|\psi_2\rangle = a(|\psi_1\rangle \otimes |\psi_2\rangle)$ , for scalar  $a$ .
4. **Distributivity over Addition.**  $(|\psi_1\rangle + |\psi_2\rangle) \otimes |\psi_3\rangle = |\psi_1\rangle \otimes |\psi_3\rangle + |\psi_2\rangle \otimes |\psi_3\rangle$ .
5. **Bilinearity.**  $(a|\psi_1\rangle + b|\psi_2\rangle) \otimes (c|\psi_3\rangle + d|\psi_4\rangle) = ac|\psi_1\rangle|\psi_3\rangle + ad|\psi_1\rangle|\psi_4\rangle + bc|\psi_2\rangle|\psi_3\rangle + bd|\psi_2\rangle|\psi_4\rangle$ .
6. **Adjoint of tensor products.**  $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$ .
7. **Action on Product States.**  $(A \otimes B)(|\psi_1\rangle \otimes |\psi_2\rangle) = A|\psi_1\rangle \otimes B|\psi_2\rangle$ .
8. **Identity behaviour (special case).**  $(I \otimes B)(|\psi_1\rangle \otimes |\psi_2\rangle) = |\psi_1\rangle \otimes B|\psi_2\rangle$ .

**9. Unitarity Preservation.** If  $U_1$  and  $U_2$  are unitary, then so is  $U_1 \otimes U_2$ .

Armed with the tensor product, let us begin with the simplest composite systems – 2-qubit states. Working with the computational basis states, we see that for  $i, j \in \{0, 1\}$ ,  $|i\rangle|j\rangle = |ij\rangle$  is a  $4 \times 1$  vector with a 1 in position corresponding to the binary number  $ij$  ( $i, j \in \{0, 1\}$ ), under the ordering  $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ , and 0 elsewhere. A general 2-qubit state can then be understood as a normalized vector in  $\mathbb{C}^4$ , or a superposition of the 4 computational basis states, given by  $|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$ , with the usual normalization condition  $|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1$ .

Measurement in this setting is also understood using the Born's rule – for example, the probability of measuring the state  $|01\rangle$  is  $|b|^2$ , while the probability of measuring 1 in the second qubit is  $|b|^2 + |d|^2$ . Furthermore, if  $|1\rangle$  is observed in the second qubit, then the resulting qubit state collapses to  $\frac{b|01\rangle + d|11\rangle}{\sqrt{|b|^2 + |d|^2}}$ . (Make note of the normalization factor.)

We are now ready to introduce what is one of the strangest, yet most fundamental pieces of the puzzle that is quantum mechanics/computing: **entanglement**. We have seen that the basis states of composite systems are essentially tensor products of the basis states of the individual, constituent systems. However, as the following exercise will tell you, this isn't necessarily true for any arbitrary state of a composite system, i.e. it may not necessarily be expressible as a tensor product of its constituent systems.

**Exercise 2.** Let  $|\psi\rangle$  be a two-qubit state. We say  $|\psi\rangle$  is separable (or product) if there exist single-qubit states  $|\phi\rangle$  and  $|\chi\rangle$  such that  $|\psi\rangle = |\phi\rangle \otimes |\chi\rangle$ . Otherwise, we say that the state is entangled.

1. Show that if  $|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$  is separable, then  $\alpha\delta = \beta\gamma$ .

**Hint:** Write  $|\phi\rangle = a|0\rangle + b|1\rangle$  and  $|\chi\rangle = c|0\rangle + d|1\rangle$ , expand the tensor product using bilinearity, and compare coefficients.

2. Use the criterion from part (a) to prove that the so-called Bell state  $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$  is entangled (not separable).

Let's work through an example to understand the difference between a separable state and an entangled one. Let  $|\psi\rangle = \frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle$ . It is not hard to see that this is a separable state, and can be expressed as  $|\psi\rangle = |+\rangle|-\rangle$ . Among other things, note that the probability of measuring  $|0\rangle$  in the first qubit is  $(1/2)^2 + (-1/2)^2 = 1/2$ ; the probability of measuring  $|1\rangle$  is the same.

Now, if the second qubit is measured, and if the outcome observed is  $|1\rangle$ , then the new state of the system is given by  $\frac{1}{\sqrt{2}}(-|01\rangle - |11\rangle)$ <sup>3</sup> Note that the probability of measuring outcome  $i$  ( $i \in \{0, 1\}$ ) in qubit 1 is still  $1/2$ .

As a contrast, suppose you have an entangled state  $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ . Here, if the second qubit is measured, and the outcome 1 is observed, then the state collapses to  $|11\rangle$ , i.e. the probability of measuring outcome  $|1\rangle$  is now 1.

We will end this lecture by briefly introducing studying a special 2-qubit gate called **CNOT**, which is typically used to generate entanglement in quantum circuits.

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<sup>3</sup>The global phase  $-1$  can of course be ignored, but has been retained for computational clarity.

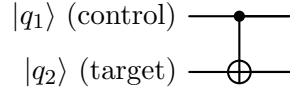


Figure 1: The  $\text{CNOT}_{12}$  (Controlled-NOT) gate flips the target qubit  $|q_2\rangle$  if and only if the control qubit  $|q_1\rangle$  is in state  $|1\rangle$ . It implements the transformation  $|a, b\rangle \rightarrow |a, a \oplus b\rangle$  where  $\oplus$  denotes addition modulo 2.

In matrix form, the  $\text{CNOT}_{12}$  gate can be expressed as

$$\text{CNOT}_{12} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

**Exercise 3.** Consider the following quantum circuit:  $\text{CNOT}_{12} \cdot (H \otimes I)|00\rangle$ .

1. Verify (using the definitions of the respective gates and the properties of the tensor product) that the output of the circuit is the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$